

Solving simultaneous equations

Basic skills

Q1. $M = \begin{pmatrix} 3 & 4 \\ 5 & -2 \end{pmatrix}$

a) $|M| = 3 \times -2 + 4 \times 5 = -6 + 20 = 14$

b) $M^{-1} = \frac{1}{|M|} \text{Adj}(M) = \frac{1}{14} \begin{pmatrix} -2 & -4 \\ -5 & 3 \end{pmatrix}$

c) $M \underline{x} = \begin{pmatrix} 5 \\ 20 \end{pmatrix} \Rightarrow \underline{x} = M^{-1} \begin{pmatrix} 5 \\ 20 \end{pmatrix}$

so $\begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{14} \begin{pmatrix} -2 & -4 \\ -5 & 3 \end{pmatrix} \begin{pmatrix} 5 \\ 20 \end{pmatrix} = \frac{1}{14} \begin{pmatrix} -100 \\ 35 \end{pmatrix}$

so $x = -\frac{100}{14}$, $y = \frac{35}{14}$

Q2. $\begin{pmatrix} 2 & 1 \\ 6 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2 \\ 7 \end{pmatrix}$

As $\begin{vmatrix} 2 & 1 \\ 6 & 3 \end{vmatrix} = 0$ no inverse so check if consistent. $2x + y = 2$

Inconsistent as $\Rightarrow 6x + 3y = 6 \neq 7$

so no solutions

Q3. $\begin{pmatrix} 1 & 4 \\ 2 & 8 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$

As $\begin{vmatrix} 1 & 4 \\ 2 & 8 \end{vmatrix} = 0$ no inverse so not unique

check if consistent $x + 4y = 2$
 $\Rightarrow 2x + 8y = 4$

so consistent so infinite solutions
 along line $x + 4y = 2$ i.e. $y = \frac{1}{2} - \frac{1}{4}x$

Competency skills

Q4.
$$\begin{aligned} 5x + 2y - z &= 8 \\ 3x - y + z &= 15 \\ x + 3z &= 18 \end{aligned}$$

a) $\begin{pmatrix} 5 & 2 & -1 \\ 3 & -1 & 1 \\ 1 & 0 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 8 \\ 15 \\ 18 \end{pmatrix}$

b)
$$\begin{vmatrix} 5 & 2 & -1 \\ 3 & -1 & 1 \\ 1 & 0 & 3 \end{vmatrix} = 5 \begin{vmatrix} -1 & 1 \\ 0 & 3 \end{vmatrix} - 2 \begin{vmatrix} 3 & 1 \\ 1 & 3 \end{vmatrix} - 1 \begin{vmatrix} 3 & -1 \\ 1 & 0 \end{vmatrix}$$

$$= 5(-3) - 2(9-1) - (0+1)$$

$$= -15 - 16 - 1 = -32$$

so invertible so unique solutions

c) For inverse need $\text{Adj}(A) = C^T$

so minors

$$|M_{1,1}| = \begin{vmatrix} -1 & 1 \\ 0 & 3 \end{vmatrix} = -3$$

$$|M_{2,3}| = \begin{vmatrix} 5 & 2 \\ 1 & 0 \end{vmatrix} = -2$$

$$|M_{1,2}| = \begin{vmatrix} 3 & 1 \\ 1 & 3 \end{vmatrix} = 8$$

$$|M_{3,1}| = \begin{vmatrix} 2 & -1 \\ -1 & 1 \end{vmatrix} = 1$$

$$|M_{1,3}| = \begin{vmatrix} 3 & -1 \\ 1 & 0 \end{vmatrix} = 1$$

$$|M_{3,2}| = \begin{vmatrix} 5 & -1 \\ 3 & 1 \end{vmatrix} = 8$$

$$|M_{2,1}| = \begin{vmatrix} 2 & -1 \\ 0 & 3 \end{vmatrix} = 6$$

$$|M_{3,3}| = \begin{vmatrix} 5 & 2 \\ 3 & -1 \end{vmatrix} = -11$$

$$|M_{2,2}| = \begin{vmatrix} 5 & -1 \\ 1 & 3 \end{vmatrix} = 16$$

$$\text{so } C = \begin{pmatrix} -3 & -8 & 1 \\ -6 & 16 & 2 \\ 1 & -8 & -11 \end{pmatrix} \quad \text{Adj}(A) = \begin{pmatrix} -3 & -6 & 1 \\ -8 & 16 & -8 \\ 1 & 2 & -11 \end{pmatrix}$$

$$\text{so } A^{-1} = \frac{1}{-32} \begin{pmatrix} -3 & -6 & 1 \\ -8 & 16 & -8 \\ 1 & 2 & -11 \end{pmatrix} = \frac{1}{32} \begin{pmatrix} 3 & 6 & -1 \\ 8 & -16 & 8 \\ -1 & -2 & 11 \end{pmatrix}$$

$$d) A\underline{x} = \underline{b} \Rightarrow \underline{x} = A^{-1}\underline{b}$$

$$\text{so } \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \frac{1}{32} \begin{pmatrix} 3 & 6 & -1 \\ 8 & -16 & 8 \\ -1 & -2 & 11 \end{pmatrix} \begin{pmatrix} 8 \\ 15 \\ 18 \end{pmatrix} = \frac{1}{32} \begin{pmatrix} 96 \\ -32 \\ 160 \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \\ 5 \end{pmatrix}$$

$$\text{so } x=3, y=-1, z=5$$

Q5. matrix $\begin{pmatrix} 6 & -1 & 2 \\ a & 2 & -4 \\ 6 & 2 & 0 \end{pmatrix}$

$$\begin{vmatrix} 6 & -1 & 2 \\ a & 2 & -4 \\ 6 & 2 & 0 \end{vmatrix} = 6 \begin{vmatrix} 2 & -4 \\ 2 & 0 \end{vmatrix} + 1 \begin{vmatrix} a & -4 \\ 6 & 0 \end{vmatrix} + 2 \begin{vmatrix} a & 2 \\ 6 & 2 \end{vmatrix}$$

$$= 6(0 + 8) + (0 + 24) + 2(2a - 12)$$

$$= 48 + 24 + 4a - 24 = 48 + 4a$$

a) $a=5, b=20$ so invertible

$$|A| = 48 + 4 \times 5 = 68$$

for cofactors (for adjugate)

$$|M_{1,1}| = \begin{vmatrix} 2 & -4 \\ 2 & 0 \end{vmatrix} = 8$$

$$|M_{2,3}| = \begin{vmatrix} 6 & -1 \\ 6 & 2 \end{vmatrix} = 18$$

$$|M_{1,2}| = \begin{vmatrix} 5 & -4 \\ 6 & 0 \end{vmatrix} = 24$$

$$|M_{3,1}| = \begin{vmatrix} -1 & 2 \\ 2 & -4 \end{vmatrix} = 0$$

$$|M_{1,3}| = \begin{vmatrix} 5 & 2 \\ 6 & 2 \end{vmatrix} = -2$$

$$|M_{3,2}| = \begin{vmatrix} 6 & 2 \\ 5 & -4 \end{vmatrix} = -34$$

$$|M_{2,1}| = \begin{vmatrix} -1 & 2 \\ 2 & 0 \end{vmatrix} = -4$$

$$|M_{3,3}| = \begin{vmatrix} 6 & -1 \\ 5 & 2 \end{vmatrix} = 17$$

$$|M_{2,2}| = \begin{vmatrix} 6 & 2 \\ 6 & 0 \end{vmatrix} = -12$$

$$\text{so } C = \begin{pmatrix} 8 & -24 & -2 \\ 4 & -12 & -18 \\ 0 & 34 & 17 \end{pmatrix}$$

$$\text{so } \text{Adj}(A) = \begin{pmatrix} 8 & 4 & 0 \\ -24 & -12 & 34 \\ -2 & -18 & 17 \end{pmatrix}$$

$$\text{so } A^{-1} = \frac{1}{|A|} \text{Adj}(A) = \frac{1}{68} \begin{pmatrix} 8 & 4 & 0 \\ -24 & -12 & 34 \\ -2 & -18 & 17 \end{pmatrix}$$

$$\begin{aligned} \text{so } \begin{pmatrix} x \\ y \\ z \end{pmatrix} &= A^{-1} \begin{pmatrix} 20 \\ -6 \\ 0 \end{pmatrix} = \frac{1}{68} \begin{pmatrix} 8 & 4 & 0 \\ -24 & -12 & 34 \\ -2 & -18 & 17 \end{pmatrix} \begin{pmatrix} 20 \\ -6 \\ 0 \end{pmatrix} \\ &= \frac{1}{68} \begin{pmatrix} 136 \\ -408 \\ 68 \end{pmatrix} = \begin{pmatrix} 2 \\ -6 \\ 1 \end{pmatrix} \end{aligned}$$

$$\text{so } x=2, y=-6, z=1$$

$$\text{b) } a=-12, b=3$$

$$|A| = 48 + 4x - 12 = 0$$

so check for consistency with $b=3$

$$6x - y + 2z = 3$$

$$-12x + 2y - 4z = -6$$

consistent so infinite solutions

with

$$6x + 2y = 0$$

$$\Rightarrow y = -3x \text{ so } 9x + 2z = 3$$

$$x = -\frac{1}{3}y \text{ and } x = \frac{1}{9}(3 - 2z)$$

$$\text{so line } x = -\frac{y}{3} = \frac{-2z + 3}{9}$$

c) $a = -12, b = 6$

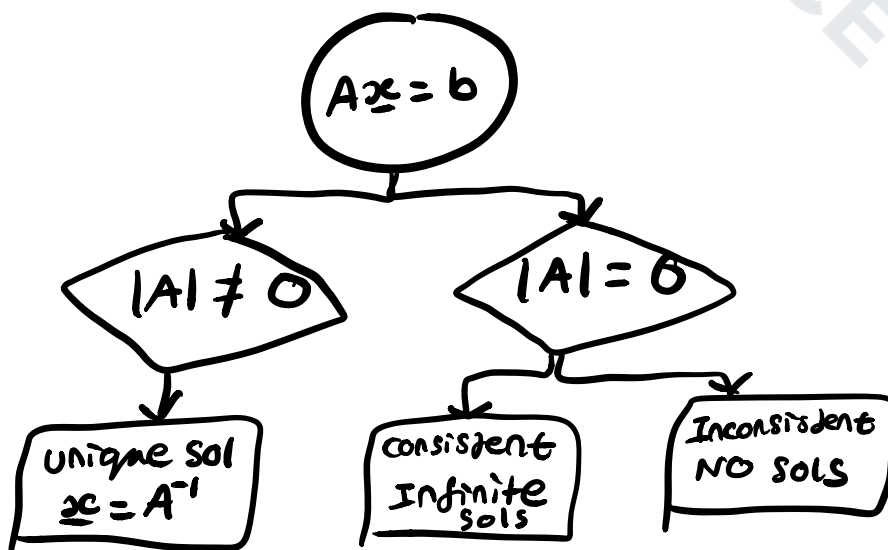
$$\Rightarrow 6x - y + 2z = 6$$

$$-12x + 2y - 4z = 6$$

inconsistent so no solutions

Advanced skills

Q6.



Q7. $\pi_1: 5x + 3y - 2z = 1$

$\pi_2: 6x + 2y + 3z = 5$

$\pi_3: 7x + y + 8z = 5$

a) For intersection

$$\begin{pmatrix} 5 & 3 & -2 \\ 6 & 2 & 3 \\ 7 & 1 & 8 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 5 \\ 5 \end{pmatrix}$$

$$\begin{vmatrix} 5 & 3 & -2 \\ 6 & 2 & 3 \\ 7 & 1 & 8 \end{vmatrix} = 5 \begin{vmatrix} 2 & 3 \\ 1 & 8 \end{vmatrix} - 3 \begin{vmatrix} 6 & 3 \\ 7 & 8 \end{vmatrix} - 2 \begin{vmatrix} 6 & 2 \\ 7 & 1 \end{vmatrix}$$

$$= 5(16 - 3) - 3(48 - 21) - 2(6 - 14)$$

$$= 5 \times 13 - 3 \times 27 - 2 \times -8 = 65 + 16 - 81$$

$$= 0$$

So not unique intersection

$$\begin{array}{r} \text{b)} \quad 9x + 3y + \frac{9}{2}z = \frac{15}{2} \\ - 5x + 3y - 2z = 1 \\ \hline \end{array}$$

$$4x + \frac{13}{2}z = \frac{13}{2}$$

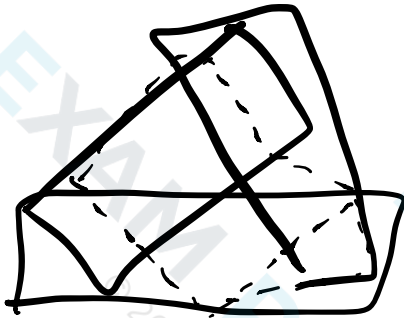
$$\begin{array}{r} - 210x + 3y + 24z = 15 \\ - 5x + 3y - 2z = 1 \\ \hline \end{array}$$

$$16x + 26z = 14$$

Now $4x + \frac{13}{2}z = \frac{13}{2}$ are inconsistent

$$16x + 26z = 14$$

no intersection at all so while
each pair intersects at line so



Prism

Q8.

$$3x + 2z = 6$$

$$x + ky + 5z = 1$$

$$-x + 2y + kz = -6$$

for unique sol $\begin{vmatrix} 3 & 0 & 2 \\ 1 & k & 5 \\ -1 & 2 & k \end{vmatrix} \neq 0$

$$\Rightarrow 3 \begin{vmatrix} k & 5 \\ 2 & k \end{vmatrix} + 2 \begin{vmatrix} 1 & k \\ -1 & 2 \end{vmatrix} \neq 0 \Rightarrow 3(k^2 - 10) + 2(2 + k) \neq 0$$

$$\Rightarrow 3k^2 + 2k - 28 \neq 0 \Rightarrow k \neq \frac{-2 \pm \sqrt{4 + 4 \times 3 \times 28}}{6}$$

$$\text{So } k \neq \frac{-1 \pm \sqrt{85}}{3} \left(\begin{array}{l} k < \frac{-1 - \sqrt{85}}{3} \\ \text{or} \\ k > \frac{-1 + \sqrt{85}}{3} \end{array} \text{ or } \frac{-1 - \sqrt{85}}{3} < k < \frac{-1 + \sqrt{85}}{3} \right)$$